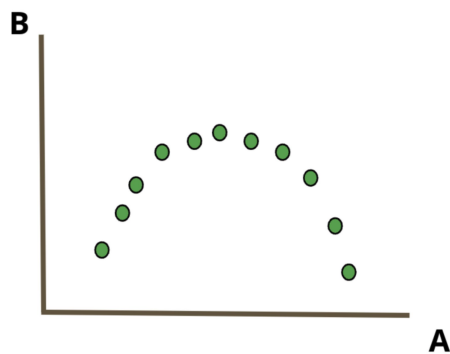


CS 506 lab 7 questions

May 2026

Questions

1. As k increases, the frobenius distance between a matrix and its rank- k approximation decreases
 - (a) True
 - (b) False
2. The rank- k approximation of an $n \times m$ matrix also has dimension $n \times m$.
 - (a) True
 - (b) False
3. What is the rank of the following dataset?



- (a) 1
 - (b) 2
 - (c) 3
 - (d) 4
4. Let A be an $m \times n$ matrix of rank r . After using SVD for feature extraction, we will still have an $m \times n$ matrix.

- (a) True
- (b) False

Solutions

1. True
2. True
3. 1
4. False